

Clinical Dataset Analysis: Gastrointestinal Diseases

This report contains structured clinical data for 30560 patients. It covers 37 variables including demographics, symptoms, lab values (CRP, Fecal Calprotectin), and diagnostic outcomes.

1. Data Dictionary & Feature Engineering

Category	Features	Description
Demographics	Age, Gender, Ethnicity, BMI	Basic patient profile and physical markers.
Clinical Markers	CRP_ESR, Fecal Calprotectin	Inflammation and biochemical indicators.
Symptoms	Abdominal Pain, Bloating, Diarrhea	Binary flags (0/1) for reported symptoms.
Diagnosis	Disease_Class	The final labeled gastrointestinal condition.

2. Global Statistical Overview

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Age	30560.0	47.22	24.92	5.0	26.0	47.0	69.0	90.0
BMI	30560.0	27.99	6.96	16.0	21.9	27.9	34.1	40.0
Body_Weight	30560.0	72.6	24.12	27.2	54.2	69.2	88.5	144.4
Family_History	30560.0	0.5	0.5	0.0	0.0	0.0	1.0	1.0
Genetic_Markers	30560.0	4.99	3.18	0.0	2.0	5.0	8.0	10.0
Microbiome_Index	30560.0	0.5	0.29	0.0	0.25	0.5	0.75	1.0
Autoimmune_Disorders	30560.0	0.5	0.5	0.0	0.0	0.0	1.0	1.0
H_Pylori_Status	30560.0	0.5	0.5	0.0	0.0	1.0	1.0	1.0
Fecal_Calprotectin	30560.0	250.93	144.79	0.0	125.0	252.0	377.0	500.0
Occult_Blood_Test	30560.0	0.5	0.5	0.0	0.0	0.0	1.0	1.0
CRP_ESR	30560.0	50.26	28.86	0.0	25.3	50.3	75.4	100.0
Endoscopy_Result	30560.0	0.5	0.5	0.0	0.0	1.0	1.0	1.0
Colonoscopy_Result	30560.0	0.49	0.5	0.0	0.0	0.0	1.0	1.0
Stool_Culture	30560.0	0.5	0.5	0.0	0.0	1.0	1.0	1.0
Food_Intolerance	30560.0	0.5	0.5	0.0	0.0	0.0	1.0	1.0
Smoking_Status	30560.0	0.5	0.5	0.0	0.0	0.0	1.0	1.0
Alcohol_Use	30560.0	0.51	0.5	0.0	0.0	1.0	1.0	1.0
Stress_Level	30560.0	5.47	2.87	1.0	3.0	5.0	8.0	10.0
Physical_Activity	30560.0	3.48	2.28	0.0	1.0	3.0	5.0	7.0
Abdominal_Pain	30560.0	0.5	0.5	0.0	0.0	0.0	1.0	1.0
Bloating	30560.0	0.5	0.5	0.0	0.0	1.0	1.0	1.0
Diarrhea	30560.0	0.5	0.5	0.0	0.0	0.0	1.0	1.0
Constipation	30560.0	0.5	0.5	0.0	0.0	1.0	1.0	1.0
Rectal_Bleeding	30560.0	0.5	0.5	0.0	0.0	1.0	1.0	1.0
Appetite_Loss	30560.0	0.5	0.5	0.0	0.0	0.0	1.0	1.0
Weight_Loss	30560.0	0.5	0.5	0.0	0.0	1.0	1.0	1.0
Bowel_Movement_Frequency	30560.0	5.03	3.16	0.0	2.0	5.0	8.0	10.0

NSAID_Use	30560.0	0.5	0.5	0.0	0.0	0.0	1.0	1.0
Antibiotic_Use	30560.0	0.5	0.5	0.0	0.0	1.0	1.0	1.0
PPI_Use	30560.0	0.51	0.5	0.0	0.0	1.0	1.0	1.0
Medications	30560.0	0.5	0.5	0.0	0.0	1.0	1.0	1.0

3. Disease Class Distribution

Disease Type	Patient Count
Abdominal cramps or pain	5243
Blood in stool	5198
Nausea or vomiting	5118
Unexplained weight loss	5047
Diarrhea or constipation	4989
Bloating	4965

4. Patient Record Samples (Detailed View)

Note: Records are split into logical feature groups for readability.

Feature Set: Age, Gender, BMI...

Age	Gender	BMI	Body_Weight	Obesity_Status	Ethnicity	Family_History	Genetic_Markers	Microbiome_Index
9	Female	19.1	44.1	Normal	Black	0	9	0.25
78	Male	38.2	121.0	Obese	White	1	10	0.27
20	Male	34.2	79.0	Obese	Black	1	6	0.83
73	Female	30.8	68.4	Obese	Asian	0	1	0.67
58	Male	26.4	66.7	Overweight	White	0	1	0.84
89	Male	34.5	63.8	Obese	White	0	9	0.85
29	Male	28.3	83.7	Overweight	Hispanic	0	4	0.47
63	Male	27.3	95.5	Overweight	Hispanic	0	5	0.6
23	Male	23.4	55.5	Normal	Hispanic	1	7	0.13
16	Female	32.2	103.2	Obese	Black	1	3	0.16
28	Male	30.6	93.7	Obese	Black	0	5	0.7
79	Male	35.7	114.4	Obese	White	0	8	0.43
24	Male	37.7	91.7	Obese	Asian	0	8	0.38
34	Male	24.3	67.0	Normal	Black	0	6	0.4
17	Female	22.2	52.0	Normal	Black	0	8	0.55
77	Female	34.2	96.5	Obese	White	1	0	0.42
81	Female	22.5	46.7	Normal	Hispanic	0	0	0.89
76	Female	36.0	73.6	Obese	Asian	1	1	0.53
22	Female	16.8	39.8	Underweight	White	0	0	0.66
58	Female	35.9	90.8	Obese	Hispanic	0	8	0.69
78	Female	34.9	92.7	Obese	Hispanic	0	3	0.23
72	Male	38.4	109.7	Obese	White	1	0	0.52
34	Male	18.2	43.2	Underweight	White	1	2	0.71
28	Male	24.0	48.4	Normal	Asian	0	0	0.86
30	Male	18.5	58.0	Normal	Other	0	5	0.35

29	Female	37.9	93.4	Obese	Black	1	1	0.02
81	Male	21.8	59.4	Normal	White	1	6	0.74
48	Male	34.1	67.8	Obese	White	0	10	0.79
17	Female	24.3	69.4	Normal	White	0	7	0.98
11	Female	37.2	108.8	Obese	Other	1	4	0.04

Feature Set: Autoimmune Disorders, H Pylori Status, Fecal Calprotectin...

Autoimmune_Disorders	H_Pylori_Status	Fecal_Calprotectin	Occult_Blood_Test	CRP_ESR	Endoscopy_Result	Colonoscopy_Result	Stool_Culture	Diet_Type
1	1	290	0	24.2	1	0	0	Vegan
1	0	390	0	22.6	1	1	0	Omnivore
0	0	264	1	57.9	0	1	1	Vegetarian
0	1	485	0	12.3	1	0	0	Pescatarian
1	0	280	0	69.6	1	1	0	Pescatarian
1	1	130	0	86.0	0	0	1	Omnivore
1	1	154	0	93.4	0	1	0	Vegan
1	1	413	0	74.9	1	1	1	Vegan
1	1	54	0	89.7	1	0	0	Vegetarian
0	0	54	1	14.3	0	1	1	Pescatarian
1	0	485	1	14.9	0	1	0	Pescatarian
0	1	362	1	96.8	1	0	1	Vegetarian
1	1	384	1	82.1	1	0	1	Vegetarian
0	0	218	0	22.3	1	0	0	Vegetarian
0	1	450	0	9.7	1	0	0	Pescatarian
1	1	361	1	19.7	1	0	0	Vegan
1	0	66	0	48.4	1	1	1	Vegan
1	1	472	1	35.1	0	0	1	Pescatarian
0	0	49	0	53.1	1	1	0	Pescatarian
1	1	491	0	17.5	1	0	1	Pescatarian
1	0	422	1	74.6	0	1	1	Vegan
0	0	237	1	15.6	1	1	1	Pescatarian

1	1	448	0	69.1	0	0	0	Vegetarian
1	0	351	0	95.1	1	0	0	Pescatarian
0	0	99	1	46.9	0	1	0	Vegetarian
0	1	98	1	28.1	1	0	1	Vegetarian
1	0	183	0	17.8	0	1	1	Vegan
1	0	136	0	38.7	1	1	1	Vegetarian
0	0	169	0	91.1	0	0	1	Omnivore
1	1	314	1	65.5	1	1	0	Vegan

Feature Set: Food Intolerance, Smoking Status, Alcohol Use...

Food_Intolerance	Smoking_Status	Alcohol_Use	Stress_Level	Physical_Activity	Abdominal_Pain	Bloating	Diarrhea	Constipation
0	0	1	9	2	0	0	0	1
1	0	0	5	0	1	0	1	0
0	1	0	4	3	0	1	0	1
1	1	0	7	6	0	0	0	1
0	1	1	3	3	0	0	0	1
0	0	1	4	4	0	0	0	0
0	0	1	3	6	1	0	1	0
0	1	0	7	7	0	0	1	0
0	1	1	6	4	1	0	0	0
1	0	0	3	0	1	1	1	0
0	0	1	9	0	1	1	0	0
0	0	1	6	2	1	0	0	1
0	1	0	9	5	0	1	0	0
0	1	1	3	3	1	1	1	0
0	0	1	5	5	1	0	1	0
0	1	0	9	4	1	1	0	1
1	0	0	9	0	0	0	0	0
0	0	0	9	2	0	1	0	1
0	1	0	9	5	0	1	0	0

1	1	1	10	5	0	1	1	0
1	1	1	2	3	1	1	1	1
1	1	0	4	0	1	1	1	0
1	1	1	9	4	0	1	0	1
0	0	0	8	0	1	1	1	0
1	1	0	4	2	0	0	0	0
1	1	0	2	0	1	1	0	0
0	1	0	6	7	0	0	1	0
1	0	0	9	5	0	1	1	1
0	0	1	9	2	1	0	1	0
0	0	1	6	5	0	1	0	0

Feature Set: Rectal Bleeding, Appetite Loss, Weight Loss...

Rectal_Bleeding	Appetite_Loss	Weight_Loss	Bowel_Habits	Bowel_Movement_Frequency	NSAID_Use	Antibiotic_Use	PPI_Use	Medications
1	1	1	Irregular	5	0	0	0	1
0	1	0	Normal	4	0	1	1	1
0	0	0	Frequent	0	1	0	0	0
0	1	0	Infrequent	2	0	1	0	1
0	1	0	Normal	4	0	0	1	1
1	1	1	Infrequent	1	0	1	0	1
0	0	1	Frequent	8	0	1	0	1
1	1	0	Frequent	1	0	1	0	0
0	0	0	Infrequent	10	1	1	0	0
1	1	1	Infrequent	3	0	0	0	1
0	0	1	Normal	10	1	0	0	0
0	1	0	Frequent	9	1	1	0	1
1	0	0	Irregular	10	0	1	1	0
0	1	0	Irregular	7	0	0	1	0
1	0	1	Infrequent	7	0	0	1	0
0	0	1	Irregular	7	0	0	1	0

1	0	0	Frequent	4	0	0	1	0
1	0	1	Frequent	2	0	0	1	0
0	1	1	Normal	7	1	1	0	0
1	0	0	Frequent	6	0	1	0	1
0	0	1	Frequent	0	1	0	0	0
1	0	1	Infrequent	2	1	1	0	1
1	0	1	Normal	7	0	0	0	0
1	1	1	Frequent	5	0	0	1	0
1	0	0	Irregular	2	0	1	0	1
0	0	1	Normal	5	0	1	0	1
0	1	0	Frequent	2	1	1	1	0
1	0	0	Normal	10	0	0	1	0
0	0	0	Frequent	6	1	1	0	0
1	0	1	Normal	9	1	0	1	1

Feature Set: Disease_Class...

Disease_Class
Unexplained weight loss
Blood in stool
Nausea or vomiting
Nausea or vomiting
Unexplained weight loss
Nausea or vomiting
Unexplained weight loss
Bloating
Blood in stool
Blood in stool
Abdominal cramps or pain
Nausea or vomiting
Nausea or vomiting

Abdominal cramps or pain
Diarrhea or constipation
Abdominal cramps or pain
Abdominal cramps or pain
Diarrhea or constipation
Abdominal cramps or pain
Nausea or vomiting
Abdominal cramps or pain
Blood in stool
Nausea or vomiting
Unexplained weight loss
Diarrhea or constipation
Blood in stool
Nausea or vomiting
Unexplained weight loss
Nausea or vomiting
Bloating